

Simulation Object Identification Module (SOIM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Governed Space: Simulation Object Identification

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Canonical Language: English (UCL™)

Governed Space

Simulation Object Identification

Canonical Definition

Simulation Object Identification Module (SOIM) defines the governance conditions under which the exact object, system, process, environment, behavior, event, interaction, mission, population, infrastructure, organization, or other target intended to be represented through simulation is formally identified, distinguished from materially different targets, connected to sufficient identity attributes, and established as a traceable Simulation Object before downstream simulation governance proceeds.

SOIM ensures that simulation does not begin from an assumed, ambiguous, substituted, overly broad, or insufficiently identifiable target.

Operational Role

SOIM establishes the first internal governance condition of the Simulation Object Standard (SOS).

Its primary question is:

What exactly is the object that this simulation claims to represent?

SOIM transforms an intended simulation target into an identifiable Simulation Object capable of being governed by subsequent SOS mechanisms.

The governed progression is:

Candidate Simulation Target → Object Identification → Object Distinction → Identity Attribute Establishment → Object Reference → Identification Status → Governed Simulation Object Identity

SOIM does not determine the Simulation Object boundary, version, configuration, or complete documentation lifecycle.

It establishes the identity anchor upon which those subsequent object-governance functions depend.

Module Operational Space

SOIM governs: candidate Simulation Object identification, Simulation Object naming, object identifiers, object type, object class, object-instance distinction, object-family distinction, physical and digital identity references, actual and hypothetical object identification, proxy-object identification, composite-object identification, constituent-object identification, object hierarchy, material identity attributes, object-reference sources, object ambiguity, duplicate identification, conflicting identity claims, object substitution detection, identity continuity, and Simulation Object identification traceability.

Module Function

SOIM exists to prevent simulation from being performed against an object whose identity is: undefined, assumed, overly generic, inconsistent, duplicated, confused with a related object, silently substituted, represented through an undisclosed proxy, or materially broader than the object actually being simulated.

Its function is not merely to assign a name.

It establishes a sufficiently precise identity relationship between:

the simulation activity

and:

the object that activity claims to represent.

Simulation Object Identification

Every governed simulation requires an identifiable target.

A Simulation Object may be: a physical asset, digital system, AI system, autonomous system, process, behavior, organization, mission, infrastructure, environment, population, event, hypothetical system, future system, or composite system.

The object need not physically exist.

It must, however, be sufficiently identifiable to preserve the meaning of subsequent simulation activity.

Identification Sufficiency

Simulation Object identification is sufficient when the available identity attributes distinguish the intended target from materially different targets that could alter: simulation representation, assumptions, model selection, input requirements, scenario design, execution, output interpretation, reality correlation, or downstream simulation relevance.

Identification sufficiency therefore depends upon the simulation context.

A broad conceptual study may tolerate broader object identity.

A high-consequence operational simulation may require significantly greater identity precision.

Object Class and Object Instance

SOIM distinguishes:

Object Class

from:

Object Instance

For example:

Autonomous Vehicle

may describe an object class.

Vehicle Platform X

may describe a narrower class.

Vehicle X · Hardware Revision 4 · Sensor Package C

may identify the materially relevant target.

SOIM requires the identification level necessary for the simulation claim being made.

Material Identity Attributes

A Material Identity Attribute is an attribute whose difference could materially alter the meaning or applicability of the simulation.

Such attributes may include: product identity, system designation, component

structure, architecture, hardware generation, software generation, mission identity, operating role, physical asset identity, population definition, geographic identity, temporal identity, or other distinguishing information.

SOIM does not require every possible attribute.

It requires the attributes necessary to prevent materially different targets from being treated as the same Simulation Object.

Object Distinction

Two Simulation Objects must remain distinguishable where their differences could materially affect simulation.

Similarity alone does not establish equivalence.

Objects are not necessarily identical merely because they share: a manufacturer, product family, system architecture, name, general purpose, physical appearance, or functional role.

The governing question is:

Would treating these objects as identical materially affect the simulation?

If yes, the distinction must remain visible.

Object Identity vs Simulation Representation

SOIM establishes a critical distinction:

Simulation Object Identity $\neq$ Simulation Representation Identity

The object being simulated and the artifact representing that object in simulation may each have their own identity.

For example:

Spacecraft Design Revision D

may be the Simulation Object.

Digital Simulation Model v8.4

may be the representation used later.

These identities must not be collapsed.

Object Identity vs Simulation Instance

SOIM also distinguishes:

Simulation Object

from:

Simulation Instance

A single Simulation Object may be used across thousands of executions.

Each execution does not become a new Simulation Object merely because the simulation is run again.

This distinction preserves continuity across repeated simulation activity.
Actual, Proposed, and Hypothetical Objects

A Simulation Object may be:

Existing

Proposed

Hypothetical

Future-State

or:

Counterfactual

Where the object does not yet exist, identity may be established through:
approved design records, architecture definitions, controlled specifications,
planning documents, configuration definitions, or another governed reference.

The absence of a physical object does not eliminate the need for object
identification.

Proxy Simulation Objects

A proxy may be used where the target object cannot be directly represented.

SOIM allows proxy use but requires explicit distinction between:

Target Simulation Object

and:

Proxy Simulation Object

Examples may include: prototype systems, predecessor systems, scaled models,
representative populations, synthetic systems, surrogate components, or
reference architectures.

SOIM establishes:

Proxy Object ≠ Target Object

The proxy relationship must remain visible.

Object Substitution

Object Substitution occurs when one object is simulated while the resulting simulation is presented as representing another object without a governed basis for that transfer.

SOIM prohibits silent substitution.

Where substitution occurs, the record should identify:

Object Intended

Object Actually Used

and:

Nature of the Relationship

Subsequent SIMULOS® governance may determine what that relationship means for simulation relevance.

Composite Simulation Object Identification

Several interacting objects may collectively form a single Simulation Object.

For a composite target, SOIM preserves:

Composite Object Identity

and:

Constituent Object Identities

where component-level differences may affect the simulation.

This allows system-of-systems simulations to remain understandable without reducing the target to one undifferentiated label.

Object Hierarchy

Simulation Object identification may operate across multiple levels:

System

↓

Subsystem

↓

Component



Subcomponent

or another appropriate hierarchy.

SOIM should preserve which level is actually being simulated.

A simulation concerning one component must not automatically acquire the identity of the complete system.

Multi-Object Simulation

A single simulation may include several independently relevant Simulation Objects.

For example:

Vehicle A

Vehicle B

Traffic Control System

Pedestrian Group

may all participate in the same simulation.

Where their identities materially affect the simulation, SOIM should preserve each separately rather than collapsing them into one generic target.

Identity Ambiguity

Simulation Object Identity Ambiguity exists when the available information does not sufficiently determine which object the simulation concerns.

This may arise from: inconsistent naming, missing identifiers, incomplete records, conflicting technical documentation, multiple candidate objects, undocumented substitutions, or insufficient distinction among configurations or systems.

Ambiguity must remain visible until sufficiently resolved.

Identification Conflict

An Identification Conflict exists where credible identity records point to materially different Simulation Objects.

For example:

one source may identify the simulated system as:

Revision A

while another identifies:

Revision B.

The conflict must not be silently resolved in favor of whichever identity produces the preferred simulation result.

No Assumed Identity Rule

SIMULOS® establishes:

Missing identity information must not be silently converted into known identity.

Where an assumption is necessary, the assumption must be explicit and transferred into the appropriate assumption-governance space.

No Silent Object Substitution Rule

A simulation performed against Object A must not be represented as simulation of Object B unless the relationship between A and B is explicitly identified and governed.

No Object Inflation Rule

The claimed Simulation Object must not be broader than the object identity actually established.

Therefore:

Component ≠ Entire System

One Variant ≠ Product Family

One Population ≠ All Populations

One Mission Configuration ≠ All Mission Configurations

unless a governed basis supports broader applicability.

No Automatic Identity Inheritance Rule

A new or related object does not automatically inherit the Simulation Object identity of another object merely because it: derives from the same design, shares the same architecture, belongs to the same product family, performs the same function, or replaces an earlier system.

Material identity continuity must be established rather than presumed.

Object Identity Continuity

Once established, Simulation Object identity should remain traceable through downstream simulation governance.

The relationship:

Simulation Object → Representation → Assumptions → Model → Inputs → Scenarios
→ Execution → Outputs

should not lose the original object reference.

SOIM therefore provides an identity anchor consumed throughout the broader SIMULOS® lifecycle.

Object Identity Change

Where the underlying target materially changes, SOIM must allow determination of whether the simulation still concerns:

the same identifiable object

or:

a materially different Simulation Object.

Detailed version and configuration governance occurs elsewhere within SOS.

SOIM governs whether fundamental identity continuity still exists.

Simulation Object Identification Status

Possible identification statuses may include:

Identified

Conditionally Identified

Partially Identified

Identification Conflicted

Identification Ambiguous

or:

Identification Undetermined

These statuses describe object-identification strength.

They do not determine simulation quality, simulation validity, or operational reliance.

Simulation Object Identification Record

SOIM should establish a persistent Simulation Object Identification Record sufficient to reconstruct the identity basis used when downstream simulation

governance began.

The record may preserve: Simulation Object Identifier, object name, object type, object class, object-instance reference, material identity attributes, actual/proposed/hypothetical status, proxy status, composite-object status, constituent-object references, identity source references, unresolved ambiguity, identification conflict, identification status, and identity traceability.

Failure Conditions

SOIM governance failure may exist where: no Simulation Object can be sufficiently identified, object identity remains materially ambiguous, materially different objects are treated as identical, an object class is represented as a specific object without sufficient basis, a proxy remains undisclosed, object substitution is concealed, multiple objects are incorrectly collapsed, composite-object components lose necessary identity, the claimed object becomes broader than the identified object, identity conflicts are ignored, or downstream simulation artifacts cannot be traced back to the object they claim to represent.

Minimum Implementation Framework

1. Identify the Candidate Simulation Target

Establish the entity, system, process, behavior, environment, event, population, mission, or other target proposed for simulation. 2. Establish Material Identity Attributes

Determine which identity attributes are necessary to distinguish the target from materially different objects. 3. Resolve or Record Identity Ambiguity

Identify conflicting, missing, proxy, substitute, composite, or uncertain identity conditions and prevent them from becoming hidden assumptions. 4. Assign Simulation Object Identification Status

Classify the strength of the object identity basis as identified, conditional, partial, conflicted, ambiguous, or undetermined. 5. Create the Simulation Object Identification Record

Preserve the object identifier, material identity attributes, references, limitations, ambiguity, status, and traceability required for downstream SOS governance.

Governance Outputs

SOIM may produce: Simulation Object Identifier, Simulation Object Identification Record, Object Type Record, Object Class Record, Object Instance Record, Material Identity Attribute Record, Object Distinction Record, Composite Simulation Object Record, Constituent Object Register, Object Hierarchy Record, Proxy Simulation Object Declaration, Object Substitution Record, Identification Conflict Record, Identity Ambiguity Record, Simulation Object Identification Status, Object Identity Continuity Record, Object Identity Change Record, and Simulation Object Identity Traceability Map.

Use Case 1 — Future Spacecraft Simulation

Scenario

A space organization intends to simulate a spacecraft before the final physical spacecraft exists.

Several design revisions are under development. Application

SOIM identifies the Simulation Object through the applicable controlled spacecraft design and establishes the material identity attributes that distinguish it from previous and alternative revisions.

The simulation remains bound to:

Spacecraft Design Revision D

rather than a generic:

Future Spacecraft

A later revision materially changes propulsion architecture.

That revision does not silently inherit the identity of the previously simulated object. Result

Simulation outputs remain attributable to the exact spacecraft design they were generated to represent, enabling later SIMULOS® governance to determine whether additional simulation is necessary after design change. Use Case 2 — Autonomous Vehicle Product Family

Scenario

An autonomous vehicle manufacturer operates several vehicles under the same commercial model name.

Different vehicles use materially different sensor packages and autonomy software generations. Application

SOIM prevents the commercial product name from functioning as the only Simulation Object identity.

The simulated vehicle is identified at the level required to distinguish the relevant system from materially different vehicles in the same family.

A simulation conducted using one sensor and autonomy configuration therefore remains anchored to that specific Simulation Object. Result

Simulation findings cannot automatically be represented as applying to every vehicle sold under the same model name.

The object identity remains sufficiently precise for subsequent boundary, version, configuration, scenario, execution, and output governance.

Architectural Position

Simulation Object Identification Module (SOIM) is the first internal governance space of the Simulation Object Standard (SOS).

It answers:

Which object?

The remaining SOS governance then establishes:

Where does that object begin and end?

Which version applies?

Which configuration applies?

and:

How must the object remain documented and traceable?

SOIM therefore provides the initial identity anchor upon which the complete Simulation Object governance layer depends.

It does not replace general identity governance under OBIDENTITY®, validation governance under VALIDOS®, integrity governance under INTEGROS®, operational reality governance under ORA™, or audit and evidence governance under ADIT®.

It governs specifically:

the identity of the object that a simulation claims to represent.

Foundational Principle

A simulation cannot be more precisely attributed than the Simulation Object to which it is explicitly identified and traceably anchored.

Canonical Closing Statement

Simulation Object Identification Module (SOIM) establishes the first internal governance mechanism of SIMULOS® by ensuring that every simulation begins with a sufficiently identifiable and distinguishable Simulation Object. Through material identity attributes, object distinction, proxy disclosure, composite-object identification, hierarchy preservation, ambiguity and conflict governance, substitution prevention, object-inflation control, identification status, and lifecycle traceability, SOIM prevents simulation outputs from being attributed to assumed, substituted, materially different, or unjustifiably broader targets. By creating a persistent Simulation Object identity anchor before boundary, version, configuration, representation, assumptions, modeling, scenario design, execution, or output interpretation proceeds, SOIM establishes the first operational foundation of governed simulation.

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